CNN Accelerator for Face Verification

COMP_ENG_452 Project

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Agenda

- Recap
- Main Design Idea
- Hardware Architecture
- Verification Method
- Performance Analysis

Recap

Requirement:
- E.g Face ID
- Focus on static image-based Face Verification

Goal:
- Reduce the inference workload on the CPU/PS side
- Maintain bit accuracy with quantized C golden model

Supported operators:
- Input-scalable 1×1 conv, 3×3 conv, 3×3/7×7 depthwise conv, residual, and PReLU

Main Design Idea

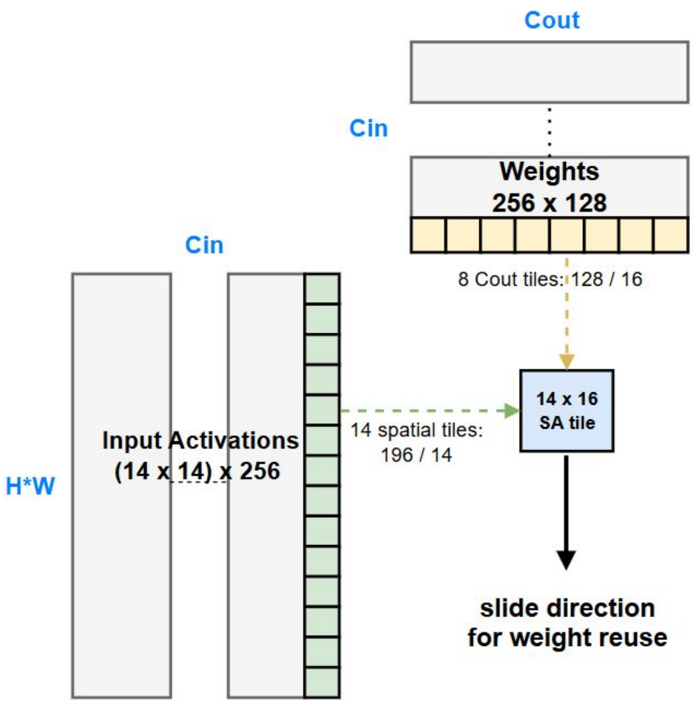
Answer Following questions:

- Why systolic array
- Why 14*16 as the array size
- What’s the quantization method
- Others

Main Design Idea

Why systolic array

- **1x1 conv accounts for 92% of the total computation**
- **Less memory bandwidth pressure (14*16*2B vs 14+16B)**



H/W: 7, 14, 28, 56
Cin: 64, 128, 256, 512
Cout: 64, 128, 256, 512

Main Design Idea

Why 14*16

- **Match MobileFaceNet layer dimensions**

Quantization method

- **symmetric weight, asymmetric activation**

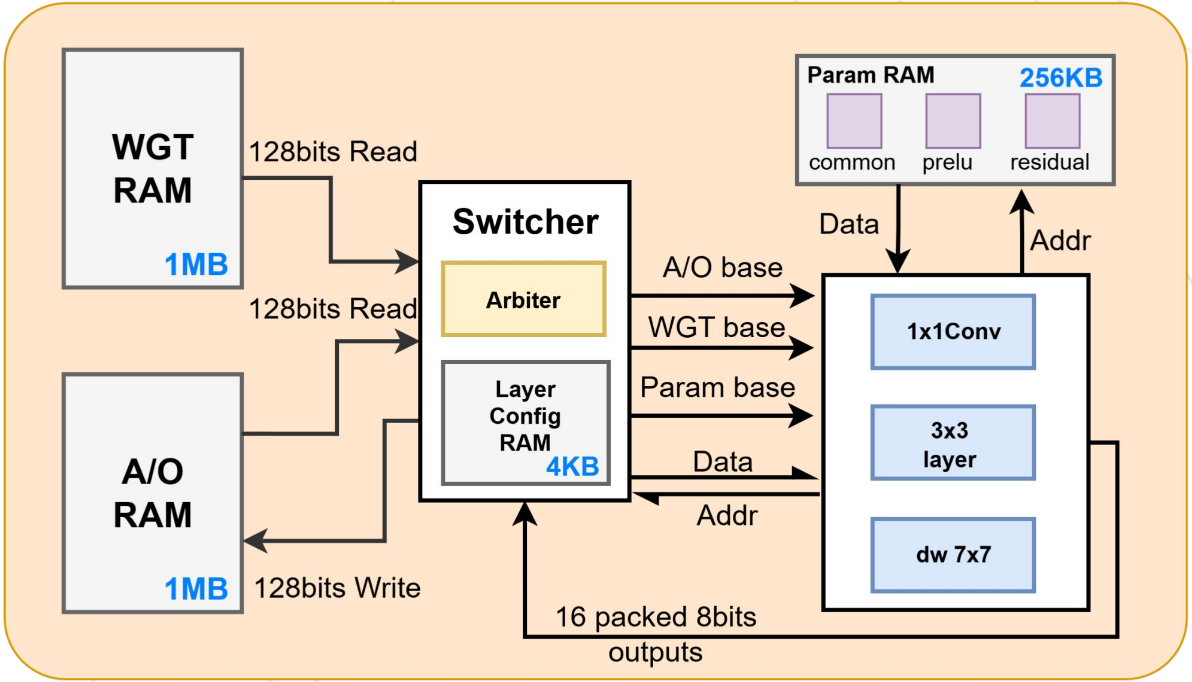
$$(act - zp_a)(wgt - \cancel{zp_w}) = act \cdot wgt - \cancel{zp_w \cdot act} - zp_a \cdot wgt + \cancel{zp_a \cdot zp_w}$$

Others: traditional 3x3 array

- **Simplifies outer controller design**
- **Effective Cin is too small, 3*3*3 and 3*3 respectively**

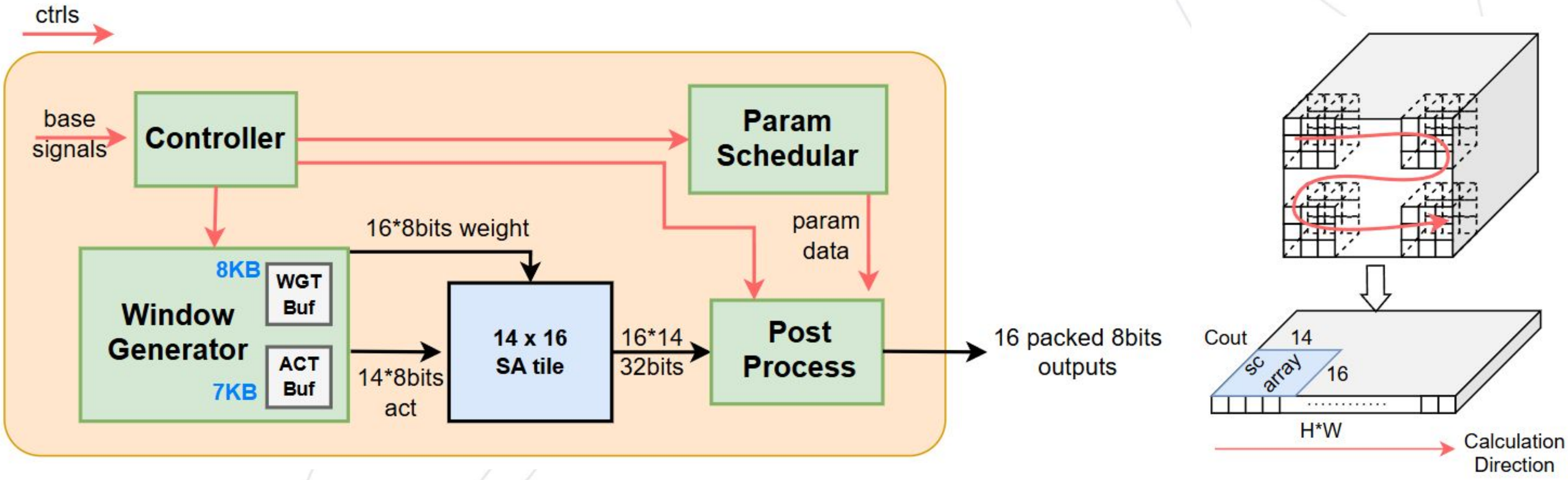
Hardware Architecture

Top Level



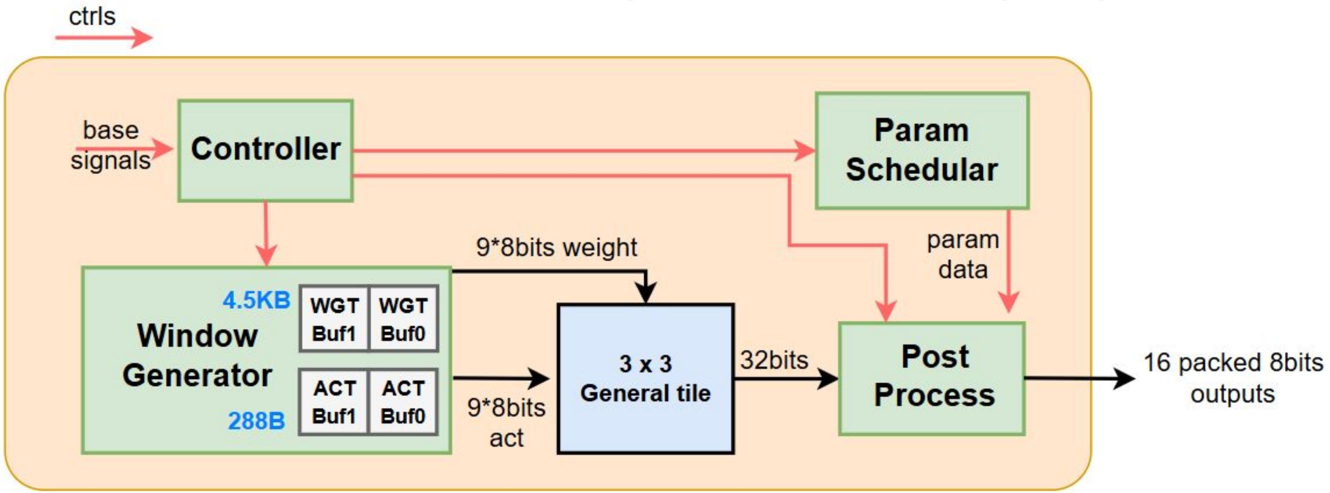
Hardware Architecture

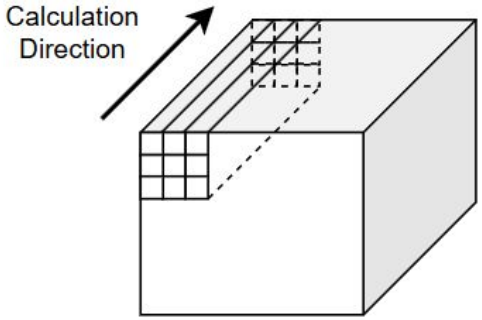
1x1 Conv



Hardware Architecture

3x3 Layer





Verification

Frontend

1. Use standard face verification benchmarks

Model	param	Quantization data	Bits	Size (MB)	LFW	CFP	AgeDb-30	CALFW	CPLFW	IJB-C	IJB-B
					Accuracy (%)					TAR@FAR 1e-4 (%)	
ResNet100	65.2M	-	FP32	261.22	99.83	98.40	98.33	96.13	93.22	96.50	95.25
		Real data	w8a8	65.31	**99.80**	**98.31**	**98.13**	**96.05**	**92.92**	**96.38**	**95.13**
		Synthetic data	w8a8	65.31	**99.80**	98.14	97.95	96.02	92.90	96.09	94.74
		Real data	w6a6	49.01	**99.55**	89.14	95.85	95.42	85.63	85.80	84.08
		Synthetic data	w6a6	49.01	99.45	**91.00**	**96.43**	**95.58**	**86.60**	**87.00**	**85.06**
ResNet50	43.6M	-	FP32	174.68	99.80	98.01	98.08	96.10	92.43	95.74	94.19
		Real data	w8a8	43.67	**99.78**	**97.70**	**98.00**	**96.00**	**92.17**	**95.66**	**94.15**
		Synthetic data	w8a8	43.67	**99.78**	97.43	97.97	95.87	92.08	95.18	93.67
		Real data	w6a6	32.77	**99.70**	95.00	97.17	**95.87**	90.17	**91.74**	**90.07**
		Synthetic data	w6a6	32.77	99.68	**95.17**	**97.43**	95.70	**90.38**	90.72	89.44
ResNet18	24.0M	-	FP32	96.22	99.67	94.47	97.13	95.70	89.73	93.56	91.64
		Real data	w8a8	24.10	**99.63**	**94.46**	97.03	**95.72**	89.48	**93.56**	**91.57**
		Synthetic data	w8a8	24.10	99.55	94.04	**97.07**	95.58	**89.53**	92.87	91.01
		Real data	w6a6	18.10	99.52	93.23	96.55	**95.58**	88.37	**93.03**	**91.08**
		Synthetic data	w6a6	18.10	**99.55**	**93.34**	**96.62**	95.32	**89.05**	92.36	90.38
MobileFaceNet	1.1M	-	FP32	4.21	99.47	91.59	95.62	95.15	87.98	90.88	88.54
		Real data	w8a8	1.10	**99.43**	**91.40**	**95.47**	**95.05**	**87.95**	**90.57**	**88.32**
		Synthetic data	w8a8	1.10	99.35	90.84	94.37	94.78	87.73	89.21	86.98
		Real data	w6a6	0.79	98.87	**87.69**	**93.03**	93.30	84.57	**83.13**	80.53
		Synthetic data	w6a6	0.79	**99.08**	87.64	91.77	**93.48**	**84.85**	82.94	**80.58**

[4]

- **LFW**: 13,233 images / 5,749 identities
- **CFP**: 7,000 images / 500 subjects,
- **AgeDB**: 12,240 images / 440 subjects, age variation
- **CALFW / CPLFW**: LFW-style 6,000-pair protocol with age/pose challenges
- **IJB-B / IJB-C:** larger-scale, more realistic benchmarks

Verification

Frontend

2. Extract Pythorch model into HW fixed-point specification
3. Compare C model with the INT8-traind Python model
4. Make the RTL design bit-accurate with C model

	LFW	AgeDB-30
QuantFace PyTorch	99.33%	95.12%
HW-spec fixed-point	99.13%	95.23%
Accuracy drop	0.20%	-0.12%

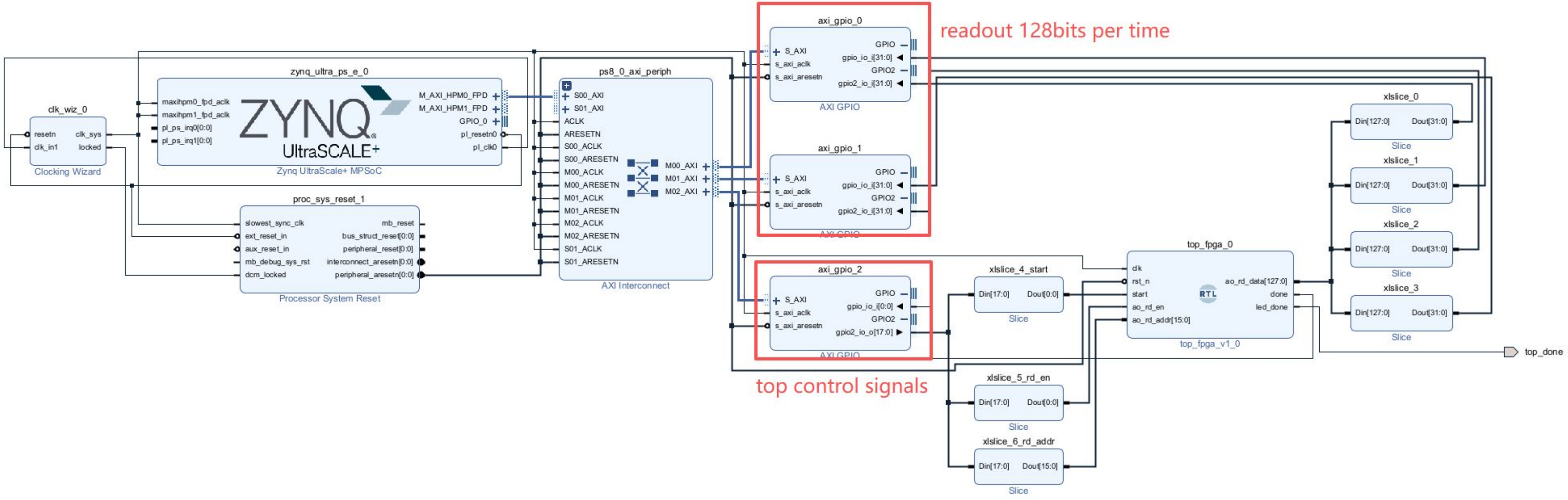
Statistical error

ps. the accuracy refers to:
(image_a, image_b, issame), 1 = same person, 0 = different person

Verification

Backend

The PS triggers the NPU execution and reads back the embedding vector upon completion.



Verification

Backend

```
40000   00000000000000000000000000000000
40001   a737e3d480127f1d50f7367f8039ec91
40002   177c7fa3e98080f9c363077f7f12d79f
40003   7f7f80807f1db90a8022ae807f967fa8
40004   5784a88080ce17e543a32a80e55ee0dc
40005   7f9256efa77fde7f80260c071e7f1f80
40006   057f6a7fff087fcc5468365a16bbcdc4
40007   80a619807fef5480c680860ace082afe
40008   4d7f1cc5a67f80717f80d07f747f809a
40009   00000000000000000000000000000000
```

Reference

Match

In [4]:

```python
# Reload PL as reset
ol.download()

print("done before start =", read_done())

# 1. start NPU
pulse_start()

ok = wait_done(20.0, 0.001)
if not ok:
    raise RuntimeError("NPU timeout")

for addr in range(40000, 40008):
    request_read(addr)
    time.sleep(0.001)
    data = read_128bit_data()
    print(addr, hex(data))
```

```
done before start = 0
done, elapsed = 0.128619 s
40000 0xa737e3d480127f1d50f7367f8039ec91
40001 0x177c7fa3e98080f9c363077f7f12d79f
40002 0x7f7f80807f1db90a8022ae807f967fa8
40003 0x5784a88080ce17e543a32a80e55ee0dc
40004 0x7f9256efa77fde7f80260c071e7f1f80
40005 0x57f6a7fff087fcc5468365a16bbcdc4
40006 0x80a619807fef5480c680860ace082afe
40007 0x4d7f1cc5a67f80717f80d07f747f809a
```

PS part readout embedding vector

Performance Analysis

Resource

Resource	Utilization	Available	Utilization %
LUT	107569	425280	25.29
LUTRAM	1308	213600	0.61
FF	166393	850560	19.56
BRAM	583	1080	53.98
DSP	421	4272	9.85
IO	1	347	0.29
BUFG	5	696	0.72
MMCM	1	8	12.50

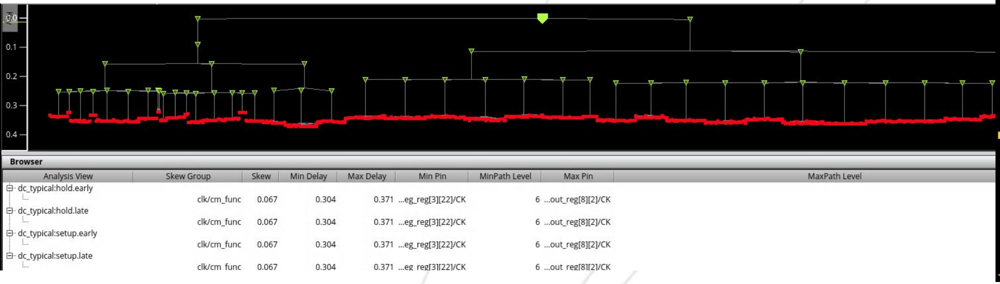
Compare

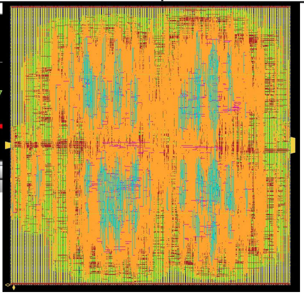
	NPU	i7-9700 3.0 GHz
Frequency	150Mhz	3.0 GHz
Total Cycles	19,175,557	-
Exe Time/image	127.8 ms	917.37 ms
Speedup	x7.18	x1

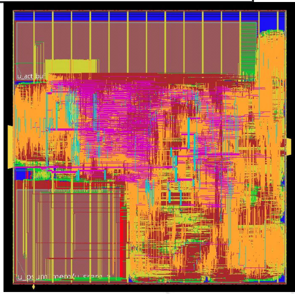
The frequency of core in RFSoC 4x2 FPGA is 500Mhz. In this case, the estimated speedup could be **x43.08**

Performance Analysis-ASIC

	v1	v2	v3
Architecture	1× 9-MAC	2× 9-MAC + CLA	**systolic array**
Memory	all FF	2 psum SRAM macro	SRAM (192b×64)
Cycles / inference	**43.16 M**	**25.77 M**	**13.77 M**
Clock	100 MHz	125 MHz	200 MHz
Inference latency	432 ms	206 ms	**69 ms**
FPS @ 720×720	2.3	4.85	**14.5**
Stdcell count	~69 K	138 K +1 SRAM	**88 K** + 2 SRAM
Core area	~285 K μm²	**479 K μm²**	**518 K μm² (720×720)**
Power	26.7 mW	11.4 mW	96.3 mW
DRC	0	0	**0**







Verification

Level	Stimulus	Delay model	Result
L1	RTL	zero-delay	50 / 50 MATCH
L2	Genus syn netlist	zero-delay	50 / 50 MATCH
L3	Innovus post-route	zero-delay	50 / 50 MATCH
L4	Innovus post-route	SDF annotated	0 violations
End-to-end embedding (RTL, 13 face images): 128 / 128 exact, MaxDiff = 0			

Run	Netlist	Delay model	Layers done	Wall-clock
`make gl`	Genus syn	zero-delay	50 / 50	~78 h
`make pnr`	Innovus post-route	zero-delay	50 / 50	~84 h
`make pnr_sdf`	Innovus post-route	SDF annotated	in progress	est. ~150 h

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1156299	ecw1531	20	0	22924	4092	1204	R	100.0	0.0	1:17.86	prog.out
1157272	mlc8919	20	0	112184	95360	1668	S	100.0	0.0	0:07.41	p_band_scan
1397938	lza6041	20	0	14948	3688	3492	R	100.0	0.0	8019:48	IR
3164394	gel8580	20	0	268296	204580	23184	R	99.7	0.1	7324:09	xmsim

Deliverables

- **Designed a hybrid CNN accelerator for MobileFaceNet inference using a 14x16 systolic array and a 3x3 standard array**
- **Supported dynamically sized 1x1 convolution, 3x3 convolution, and depthwise convolution operators**
- **NPU runs at 150 MHz on FPGA, 200MHz on ASIC, and gain x7.18 and x13.32 speedup respectively**
- **Pass FPGA & Innovus SDF backend verification.**

References

- [1] **Y.-H. Chen, T. Krishna, J. S. Emer, and V. Sze, "Eyeriss: An energy-efficient reconfigurable accelerator for deep convolutional neural networks," IEEE Journal of Solid-State Circuits, vol. 52, no. 1,pp. 127–138, 2017. doi: 10.1109/JSSC.2016.2616357.**
- [2] A. George, C. Ecabert, H. O. Shahreza, K. Kotwal, and S. Marcel, "Edgeface: Efficient face recognition model for edge devices," IEEE Transactions on Biometrics, Behavior, and Identity Science, vol. 6, no. 2, pp. 158–168, 2024. doi: 10.1109/TBIOM.2024.3352164.
- [3] S. Chen, Y. Liu, X. Gao, and Z. Han, Mobilefacenets: Efficient cnns for accurate real-time face verification on mobile devices, 2018. arXiv: 1804 . 07573 [cs.CV]. [Online]. Available: https : / / arxiv.org/abs/1804.07573.
- [4] Fadi Boutros, Naser Damer, and Arjan Kuijper. Quantface: Towards lightweight face recognition by synthetic data low-bit quantization, 2022.